

Power + Integrated-Gain Follow-Up

Executive Summary

This follow-up is post-hoc and does not overturn the pre-registered weak_or_ambiguous call. It only informs the forward decision about whether to spend GPU on a real conditional-seed ensemble.

The crude ensemble is biased structural perturbation, not a posterior; a positive result licenses testing the real ensemble, not a claim that uncertainty-aware selection works.

The enriched set is paper-pipeline-selected, not an independent gold standard; retention is a calibration diagnostic.

Forward call: do_not_build_strict_call_stands.

This does not change the registered pre-test call. The registered gate remains weak_or_ambiguous.

Pre-Registered Rules for This Follow-Up

Analysis A was declared underpowered if $b + c \leq 30$ and the exact-CI MDE exceeded the log2-rate extrapolated expected effect. It was declared an informative null if $b + c > 30$, MDE was smaller than the expected effect, and the point estimate was approximately zero.

Analysis B used a single shared lambda across all rates. The primary statistic is IG_retention(lambda) under equal_log weights, where lambda maximizes integrated retention gain over positive lambdas only. A build call requires positive mild IG with CI excluding zero, recall not below zero beyond its bootstrap half-width, and mild_no_full matching the sign at the same lambda.

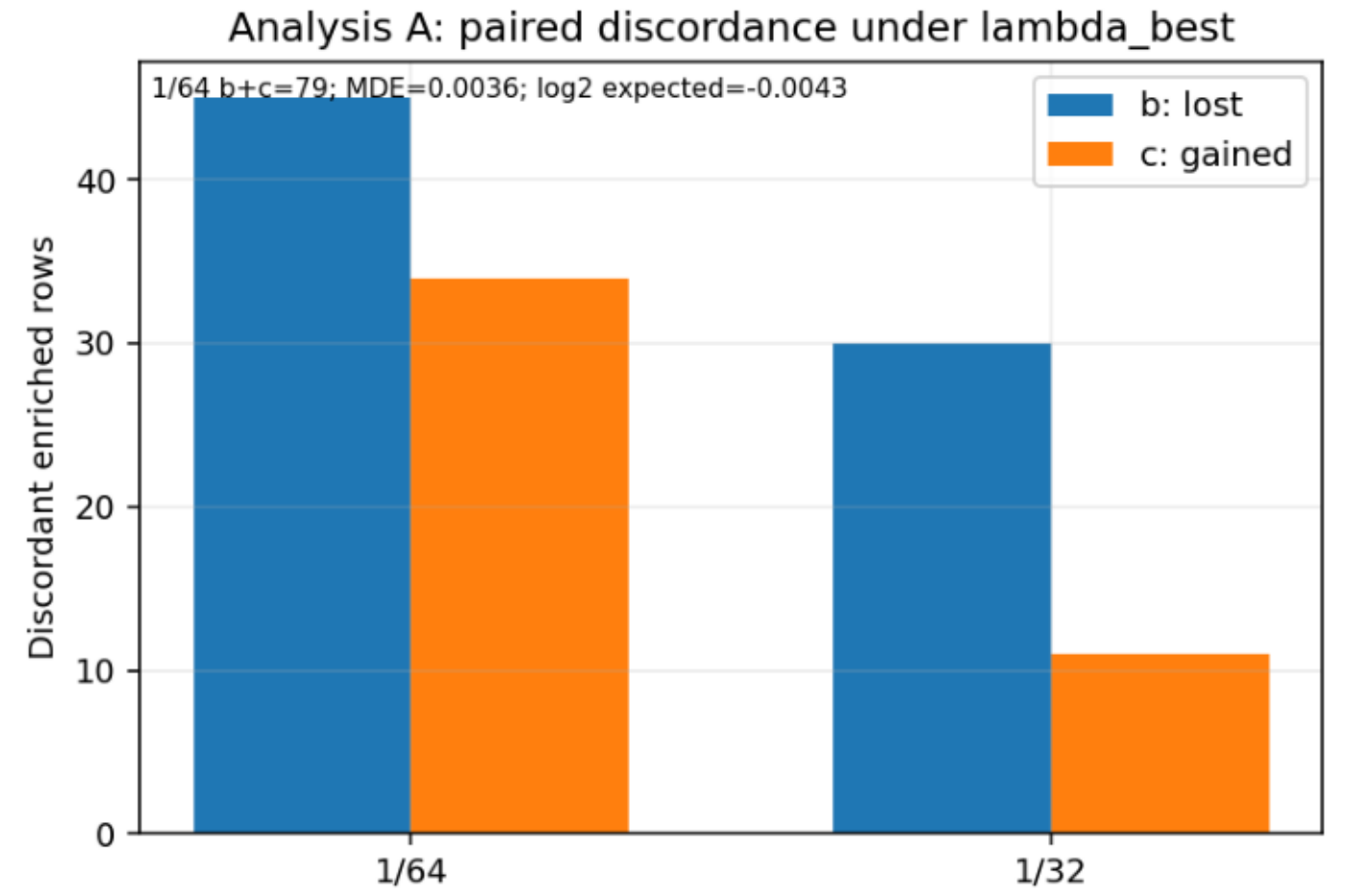
Analysis A: 1/64 Power and Discordance

mild best positive lambda from the pre-test at 1/64: 0.25.

Table

rate	b lost	c gained	b+c	diff	exact 95% CI	MDE
1/64	45	34	79	-0.0022	[-0.0057, 0.0015]	0.0036
1/32	30	11	41	-0.0038	[-0.0059, -0.0012]	0.0024

At 1/64, b+c=79, MDE=0.0036, and the primary log2-rate expected effect is -0.0043. Analysis A status: indeterminate.

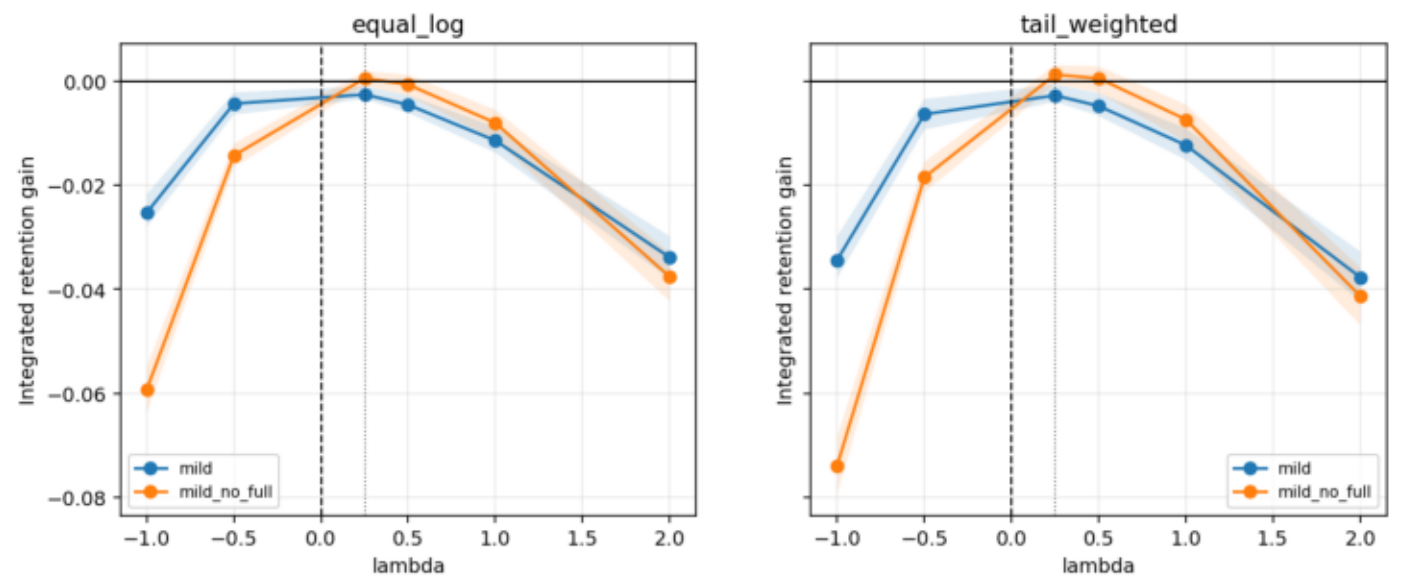


Table

ensemble	weighting	lambda*	IG retention	95% CI	IG recall	95% CI
mild	equal_log	0.25	-0.0026	[-0.0035, -0.0010]	0.0038	[0.0019, 0.0060]
mild	tail_weighted	0.25	-0.0028	[-0.0040, -0.0007]	0.0024	[0.0002, 0.0051]
mild_no_full	equal_log	0.25	0.0005	[-0.0009, 0.0019]	0.0118	[0.0093, 0.0140]
mild_no_full	tail_weighted	0.25	0.0012	[-0.0006, 0.0030]	0.0108	[0.0075, 0.0134]
mid	equal_log	0.25	-0.0027	[n/a, n/a]	-0.0156	[n/a, n/a]
mid	tail_weighted	0.25	-0.0027	[n/a, n/a]	-0.0154	[n/a, n/a]
wide	equal_log	0.5	0.0101	[n/a, n/a]	0.0333	[n/a, n/a]
wide	tail_weighted	0.5	0.0131	[n/a, n/a]	0.0336	[n/a, n/a]

Primary mild equal-log lambda* is 0.25 with retention IG -0.0026 and recall IG 0.0038. Recall tolerance is the bootstrap half-width 0.0020. mild_no_full at the same lambda has equal-log retention IG 0.0005.

Integrated retention gain curves



Combined Decision Table

Integrated gain vs recall guard

Table

Analysis A (1/64)	Analysis B (mild IG)	mild_no_full IG sign	Forward call
indeterminate	negative	does not match	do_not_build_strict_call_stands

Interpretation

The follow-up does not justify GPU spend for the real ensemble at this depth. The strict call stands.

Reproducibility

- Run hash: 6e52c70564ea1efaf07f6da0d0a780b0145648325abf02329577ff5346bb8353
- Follow-up provenance: results/pretest_followup_provenance.json
- Bootstrap resamples pure-C4 and enriched rows separately and recomputes pure-C4 thresholds within each replicate.
- Analysis B uses one lambda shared across all rates; there is no per-rate argmax.

